

CS-1180 Lab Problem 3: Grade Calculator - Conditionals

PURPOSE: To test the ability to write conditionals

DIRECTIONS: For this lab problem, you will continue modifying your Grade Calculator code. Each time you prompt the user for their grade for a specific assignment, write a conditional (if/else) statement that verifies that the user entered the correct data type. (Hint: use the “has” methods of the Scanner class to check input type). If the type is correct, then also check if the input is in a valid numerical range for a grade (0.0 to 100.0, inclusive). If at any point invalid or out of range input is given, print an appropriate error message conveying what was wrong and then terminate the program with the command `return; .`

Additionally, modify the prompt for the student’s desired grade to ask for a single character instead of a number. Write a conditional cascade (if/else if.../else) or switch statement which sets the desired grade variable from earlier assignments to the correct value depending upon the user’s input (A-> 90.0, B -> 80.0, C -> 70.0, D -> 60.0, F -> 0.0).

If the user did not enter one of the valid choices, treat the situation as if the user had entered ‘A’. Make sure to modify your print statement concerning the target Final exam grade to print the letter grade the user entered instead of the percent like this:

```
To get a A for the class, you need to get a 87.94% for your final exam.
```

NOTE: The conditionals should be case insensitive; i.e., an input of either ‘A’ or ‘a’ should be interpreted in the same way.

NOTE: `Scanner` does not have a `nextChar` method for extracting a single character from input. How will you get around that?

EXAMPLE: Running your program should look something like this. Skipped lines here indicate where the program was restarted. Depending on how your terminal or IDE is configured, the beginning of a program may look different, but the output should be similar.

```
What is your grade for zyBooks?
> grade
ERROR: Invalid input: Non-numerical. Terminating program.

What is your grade for zyBooks?
> 97.3
You have a 97.30% grade for zyBooks, which is 10% of your final grade.
What is your grade for attendance?
> 573
ERROR: Invalid input: Out of range. Terminating program.

What is your grade for zyBooks?
> 97.3
You have a 97.30% grade for zyBooks, which is 10% of your final grade.
What is your grade for attendance?
> 100
You have a 100.00% grade for attendance, which is 5% of your final grade.
```

What is your grade for lab?

> 89.72

You have a 89.72% grade for lab, which is 42% of your final grade.

What is your grade for the quizzes?

> 86.24

You have a 86.24% grade for the quizzes, which is 10% of your final grade.

What is your grade for the midterm?

> 87.5

You have a 87.50% grade for the midterm, which is 13% of your final grade.

Your grade so far is 90.51%.

What is your desired grade for the class?

> a

To get a A for the class, you need to get a 86.68% for your final exam.

RUBRIC:

(3 pts) **Base Functionality**

[1] Program checks for the correct data type for each input

[1] Program checks that entered grades are in the correct range

[1] Program correctly determines the target Final exam grade based on the desired letter grade

(1 pts) **Style**

[1] The code is meaningfully commented

IMPORTANT NOTES:

- Submissions that do not compile will receive a zero
- Submissions with improperly cited AI will receive a zero and an academic integrity violation
- Submissions that are partially or fully copied from another submission will receive a zero and an academic integrity violation